

Problem 1.

(i) Let $\langle s_1, s_2, \dots, s_n \rangle$ be the generators of M_1 .

$$0 \rightarrow M_1 \xrightarrow{\varphi} M \xrightarrow{\psi} M_2 \rightarrow 0$$

We know that $\ker \psi = \text{im} \varphi$. Therefore we can generate $\ker \psi$ with $\varphi(s_1), \varphi(s_2), \dots, \varphi(s_n)$. Now since ψ is surjective, we have that:

$$\frac{M}{\ker \psi} = M_2$$

Since M_2 is finitely generated, then $M/\ker \psi$ is also finitely generated. Therefore the set of cosets is finitely generated. Let the generators be $\langle \bar{t}_1, \bar{t}_2, \dots, \bar{t}_k \rangle$. We argue that $\langle \varphi(s_1), \dots, \varphi(s_n), t_1, \dots, t_k \rangle$ generates M . Let $m \in M$. m belongs to one of cosets, thus we can write:

$$\begin{aligned} m + \ker \psi &= r_1 \bar{t}_1 + \dots + r_k \bar{t}_k + \ker \psi \\ \implies m - r_1 t_1 - \dots - r_k t_k &= i \end{aligned}$$

for some i in $\ker \psi$. Since $\ker \psi = \text{im} \varphi$, then we can write:

$$i = u_1 \varphi(s_1) + \dots + u_n \varphi(s_n)$$

This gives us that m is generated by the generator we introduced:

$$m = r_1 t_1 + \dots + r_k t_k + u_1 \varphi(s_1) + \dots + u_n \varphi(s_n)$$

Thus M is finitely generated.

However it is not necessarily true that if M is finitely generated, then M_1 and M_2 are finitely generated as well. An example of this could be the R -modules below:

$$0 \rightarrow R[x_1, x_2, \dots] \xrightarrow{\sigma_0} R \xrightarrow{id} R \rightarrow 0$$

Where R has 1, and σ_0 is valuation at 0 for all x_i s.

(ii)

Problem 2.

(i) Consider Φ :

$$\Phi : \text{Hom}_R\left(\bigoplus_{i \in I} M_i, N\right) \rightarrow \bigotimes_{i \in I} \text{Hom}_R(M_i, N)$$

Then for each $\sigma \in \text{Hom}_R(\bigoplus_{i \in I} M_i, N)$, we have that for any $m_i \in M_i$ for all $i \in I$, there exists some $n \in N$ such that:

$$\sigma(m_1, m_2, \dots) = n$$

Now define $\sigma_i = \sigma(0, 0, \dots, 0, m_i, 0, 0, \dots)$. Meaning it only one nonzero entry in i th place. Now we map:

$$\Phi : \sigma \mapsto \bigotimes_{i \in I} \sigma_i$$

Now it is easy to see that for any $\sigma, \delta \in \text{Hom}_R(\bigoplus_{i \in I} M_i, N)$, we have:

$$\Phi(\sigma + \delta) = \bigotimes_{i \in I} (\sigma + \delta)_i = \bigotimes_{i \in I} (\sigma_i + \delta_i) = \bigotimes_{i \in I} \sigma_i + \bigotimes_{i \in I} \delta_i = \Phi(\sigma) + \Phi(\delta)$$

Thus Φ is a homomorphism. Now to see that it is surjective as well suppose $\bigotimes_{i \in I} \psi_i \in \bigotimes_{i \in I} \text{Hom}_R(M_i, N)$, then let us introduce $\psi \in \text{Hom}_R(\bigotimes_{i \in I} M_i, N)$ such that for any $\bigotimes_{i \in I} m_i$ we have:

$$\psi(\bigotimes_{i \in I} m_i) = \sum_{i \in I} \psi_i(m_i)$$

This proves the surjectivity, and thus these are isomorphic as groups.

(ii)

Problem 3.

(i)

Problem 4.

(i)

Problem 5.

Problem 6.

(i) We need to introduce a homomorphism from V to $U \oplus W$.

$$\begin{array}{ccccccccc} 0 & \longrightarrow & U & \xrightarrow{\varphi} & V & \xrightarrow{\sigma} & W & \longrightarrow & 0 \\ & & \downarrow id_U & & \downarrow & & \downarrow id_W & & \\ 0 & \longrightarrow & U & \xrightarrow{p_1} & U \oplus W & \xrightarrow{p_2} & W & \longrightarrow & 0 \end{array}$$

by exactness in the sequence we know that:

$$\ker(\varphi) = 0 \quad \text{Im}(\varphi) = \ker(\sigma) \quad \text{Im}(\sigma) = W$$

Each of these homomorphisms are uniquely identified with how they behave on a the basis. Define:

$$\begin{aligned} \psi : V &\rightarrow U \oplus W \\ v &\mapsto \langle \varphi^{-1}(v), \sigma(v) \rangle \end{aligned}$$

Since $\ker(\varphi) = 0$, then φ^{-1} is well defined. To show that This means that for any element in $v \in V$, there exist $f_1, f_2, \dots, f_n \in \mathbb{F}$ such that:

$$v = f_1 v_1 + f_2 v_2 + \dots + f_n v_n$$